

SCSE3040: Machine Learning Operations (MLOps)

Practical 05 — Settings in a File, Bugs Caught by a Robot (Laboratory Submission)

1. Student & Laboratory Session Details

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Course:	SCSE3040 — Machine Learning Operations	Practical:	P05 (L08 / CO3)

2. Repository Link

Full GitHub Repository URL: <https://github.com/div339175/P05-config-tests>

3. Contents of config.yaml

```
# Settings for the delivery-time model.
# Change values here, never in the code.

data:
  path: ../data/delivery_times.csv
  features:
    - distance_km
    - prep_time_min
    - traffic_level
    - rain
  target: delivery_min

split:
  test_size: 0.2
  seed: 42

model:
  name: linear_regression
```

4. Terminal Output (pytest Test Suite Execution)

Terminal session running pytest inside the work/ directory showing all unit and behavioural tests passing:

```
divakarmaurya — zsh — 88x26

divakarmaurya@MacBook-Pro work % pytest -v
===== test session starts =====
platform darwin -- Python 3.13.5, pytest-8.3.4, pluggy-1.5.0 -- /opt/anaconda3/bin/python
cachedir: .pytest_cache
rootdir: /Users/divakarmaurya/Downloads/P05-config-tests/work
plugins: anyio-4.7.0
collecting ... collected 10 items

test_model.py::test_rain_never_makes_delivery_faster PASSED [ 10%]
test_model.py::test_further_is_never_faster PASSED [ 20%]
test_model.py::test_error_is_better_than_guessing PASSED [ 30%]
test_model.py::test_heavier_traffic_is_never_faster PASSED [ 40%]
test_orders.py::test_minutes_per_km_simple PASSED [ 50%]
test_orders.py::test_minutes_per_km_rejects_zero_distance PASSED [ 60%]
test_orders.py::test_good_order_is_accepted PASSED [ 70%]
test_orders.py::test_bad_traffic_level_is_rejected PASSED [ 80%]
test_speed.py::test_slow_delivery PASSED [ 90%]
test_speed.py::test_negative_distance_raises PASSED [100%]

===== 10 passed in 0.02s =====
divakarmaurya@MacBook-Pro work %
```

5. Model Performance Comparison Before & After Refactor

Comparison: Before the refactor (hardcoded values in script), the package produced a test Mean Absolute Error (MAE) of **1.92 minutes**, and after refactoring settings into `config.yaml`, it produces identically **1.92 minutes** (and **2.18 minutes** when constrained to `max_rows: 300` in Task T1).

6. AI Use Disclosure

Disclosure: None